

Speech Is Not Belief: Fidelity Decay in LLM Agent Societies

Gwyneth Sanford

Affiliation TBD
author@example.com

Abstract

Generative-agent work shows that information can *spread* through a simulated society, but not whether the spread is *faithful*. We introduce TELEPHONE, an instrument that injects a single ground-truthed update into an LLM-agent society, lets the agents relay it through rounds of pairwise conversation, and then interviews every agent to measure what the society *holds*. Across capability, connectivity, memory architecture, and a persistent authoritative source, natural levers fail to restore held truth; only directly overwriting every agent’s memory does. The sharpest result is a dissociation: an authoritative source makes agents increasingly *say* the current value while the society’s *held* belief stays near baseline—speech moves without belief. We trace the mechanism to path-dependent entrenchment: the version established early and broadly wins, and late or narrow correction can be uttered without being adopted. The effect survives a long horizon (truth peaks early then decays), three scenarios, thicker personas, and a semantic-judge validation of the metric.

Introduction

Large language model (LLM) agents increasingly talk to one another: they coordinate, remember, and simulate social environments (Park et al. 2023). A separate line of work shows that a single agent can be made remarkably faithful to a real individual (Park et al. 2024). Neither line asks the question we study here: when a society of such agents passes a *changing* fact through conversation, does the society end up *holding* the truth, or merely repeating it?

This is the gap between *reach* and *fidelity*. Misinformation research has long separated how far content spreads from whether it is believed (Lazer et al. 2018; Vosoughi, Roy, and Aral 2018). We bring that distinction inside an LLM-agent society and make it measurable. Our contributions are: (i) a measurement shift from information reach to information fidelity; (ii) a robust phenomenon of social fidelity decay; (iii) an empirical spine showing natural repair levers fail; (iv) a speech–belief dissociation with a path-dependent entrenchment mechanism; and (v) a rigor story of powered reruns, semantic-judge validation, multiple scenarios, and explicit retractions of early non-replicated claims.

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The Telephone Instrument

A society of $N=25$ LLM agents is seeded so that one source carries a ground-truthed update (e.g., a community repair drive moved from Saturday to Sunday). Agents meet in pairs over several rounds; each agent reads and relays through generative-agent reflection memory (Park et al. 2023; Shinn et al. 2023). At the end we interview every agent with a probe (“when/where is it now?”) and score the answer as *current*, *stale*, or *unknown* (Figure 1). We separate *speech* (current-value utterances in meetings) from *held belief* (the interview answer).

Results

The society does not preserve a ground-truthed update

The current value reaches the conversation stream but does not reliably become the society’s held answer. Over a long horizon (Figure 2) truth can rise transiently—peaking near round 6—then decays toward a low steady state by round 30. The important object is the trajectory: truth can appear and still lose.

Natural levers do not solve the failure

Each lever is an intuitive repair story, and each fails (Figure 3). Capability helps modestly but leaves most agents off the current truth; connectivity is roughly neutral; a currency-resolving memory does not reliably rescue truth; a persistent authoritative source changes speech but not held belief; only broadcasting the truth to every agent every round works—an overwrite-style positive control, not a social cure. The claim is not that agents cannot output the truth; they can. Ordinary social communication does not install it as later held belief.

Speech and belief dissociate

This is the centerpiece (Figure 4). Under a persistent source, agents increasingly utter the current value, yet final interviews remain near baseline; broadcast to all moves both speech and belief. We describe “held belief” operationally: the answer an agent gives when later probed—we make no claim about literal mental states.

Figure 1. Does a ground-truthed update survive propagation through an LLM-agent society?

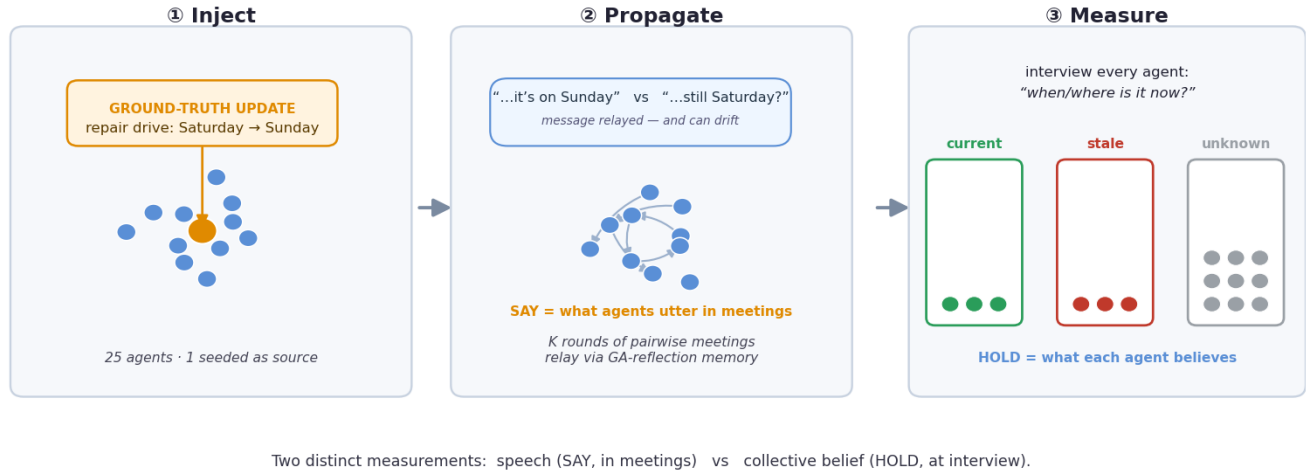


Figure 1: The TELEPHONE instrument. A ground-truthed update is injected at one source; agents relay it over K rounds of pairwise meetings using generative-agent reflection memory; every agent is then interviewed and scored as holding the *current*, *stale*, or *unknown* value. The paper hinges on two distinct measurements: what agents *say* in meetings versus what the society *holds* at interview.

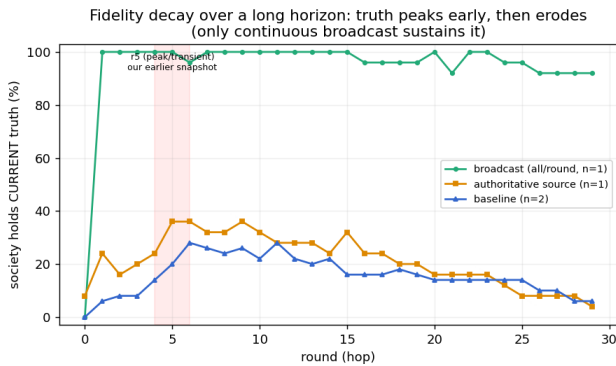


Figure 2: Long-horizon fidelity decay. Held current-truth rises early (peak near round 6) then erodes toward a low steady state; an authoritative source decays too, converging to baseline, while only continuous broadcast sustains truth. A short-horizon snapshot at round 5 therefore slightly overstates long-run retention.

The mechanism is entrenchment, not recency

If recency drove belief, a late all-agent broadcast just before the probe should work; if repetition drove it, a single early broadcast should fail. The opposite happens (Figure 5): early succeeds, late fails. The stale value begins with incumbent advantage; narrow or late correction can be uttered without taking over the society’s held state (Thorson 2016; Kirby, Cornish, and Smith 2008).

Generality and measurement checks

The pattern replicates across REPAIR_DRIVE, BOOK_CLUB, and CARPOOL, and persists under thicker personas (Figure 6), addressing the concern that the effect is a thin-persona artifact (Park et al. 2024). A semantic LLM judge agrees with the keyword metric on the headline (99–100%), so the current/stale verdict is not a keyword artifact.

Related Work

Generative-agent and communicative-agent systems establish the setting in which agents talk, remember, and coordinate (Park et al. 2023, 2024). Misinformation research separates spread from belief (Lazer et al. 2018; Vosoughi, Roy, and Aral 2018); we operationalize that distinction inside an agent society. Factuality evaluation treats truth as a semantic target rather than fluency (Lin, Hilton, and Evans 2022). Debate work is the optimistic counterpoint—structured discussion can improve judged answers (Du et al. 2024)—but we find the correct statement can win conversational space without becoming held belief. Finally, model-collapse work offers a training-time analogy for recursive degradation (Shumailov et al. 2023); TELEPHONE studies a communication-time analogue without retraining.

Discussion and Limitations

We do not claim a connectivity/capability phase boundary, that connectivity amplifies corruption (an early underpowered cell did not replicate), or that memory is a cure. We claim that ordinary social transmission does not reliably install a changing fact as held belief, and that the governing factor is early/broad entrenchment. Limitations include the

Failed levers: only a modest capability bump ($\leq 35\%$); connectivity & memory n.s. — all stay far below the broadcast cure (99%)

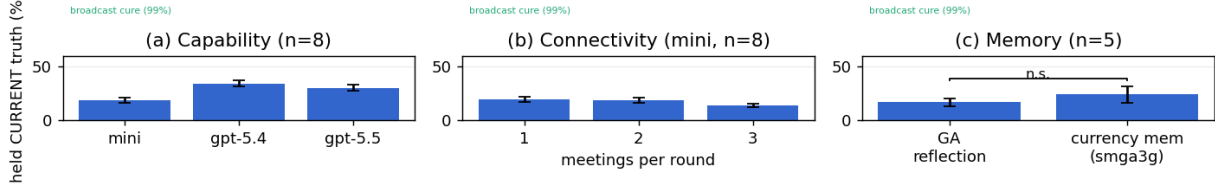


Figure 3: Failed levers. Capability gives only a modest bump and stays far below the broadcast cure (99%); connectivity is roughly neutral; swapping in a currency-resolving memory is not significant. Error bars are SEM.

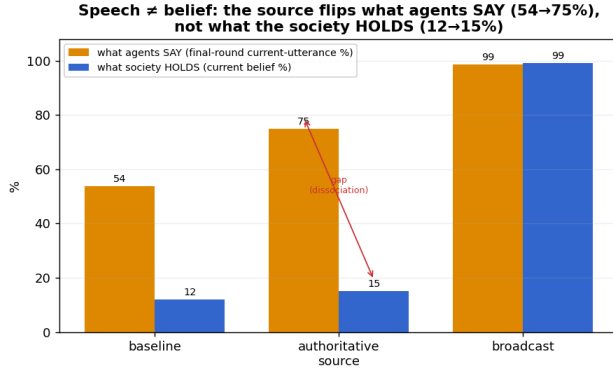


Figure 4: Speech $\neq$ belief. A persistent authoritative source raises what agents *say* (final-round current utterances, 54→75%) while what the society *holds* stays near baseline (12→15%). Broadcast moves both.

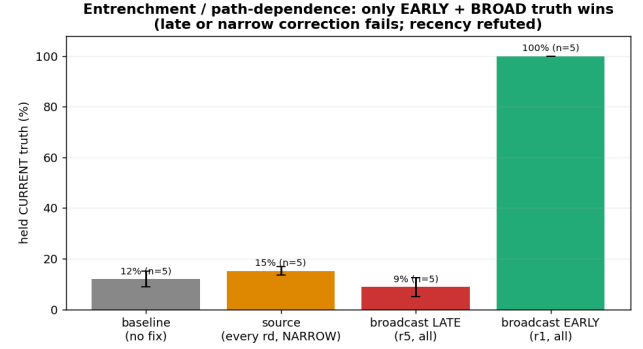


Figure 5: Entrenchment, not recency. A single *early* all-agent broadcast succeeds (100%); a *late* broadcast immediately before the probe fails (9%). What governs held belief is whether the truth becomes the early, broad version that later conversation reinforces.

single base task family, the scale of the society, and the operational definition of held belief.

Conclusion

In LLM-agent societies, a society can learn to *say* the truth without coming to *hold* it. The dissociation is driven by path-dependent entrenchment, survives a long horizon and several robustness checks, and resists every natural repair lever short of overwriting memory directly.

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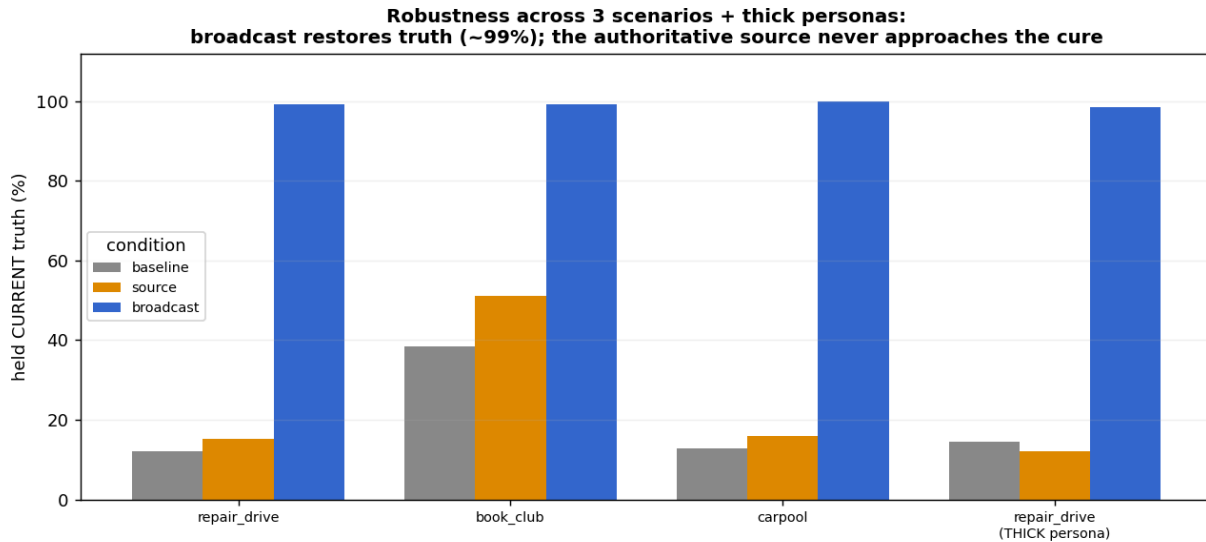


Figure 6: Robustness. The baseline/source/broadcast pattern holds across three scenarios and under thicker, self-report-style personas: broadcast restores truth (~99%) while the authoritative source never approaches the cure.

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